第一周 - D_4 Group Cayley Group representation

大綱脈絡

1. Define the elements of the group D_4.

2. Construct the Cayley graph of D_4, where:
   - Each group element corresponds to a vertex.
   - Generators correspond to edges.
   - Edges are defined via left multiplication.

3. Assign an index to each group element and use this indexing to construct:
   - Adjacency matrix.
   - Degree matrix.
   - Graph Laplacian.

4. Compute the eigenvectors of the Laplacian matrix. Select the eigenvectors corresponding to the four smallest eigenvalues. Since the smallest eigenvalue is 0, we discard its eigenvector and use the next three eigenvectors to obtain an embedding in $\mathbb{R}^3$.

5. Visualize the resulting embedding in the low-dimensional space.

6. Add a face structure to the Cayley graph by selecting relator-induced cycles.

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https://colab.research.google.com/drive/1_f7YCLz9UT6IwYlArkb8cURHYKkVVaJU?usp=sharing

Constructing Surfaces from Cayley Graphs

Project Goal

Given a Cayley graph

$$\Gamma = \mathrm{Cay}(G, S),$$

we want to interpret it as the **1-skeleton of a closed surface**.

Our goal is not only to construct a surface, but to construct one where the group G **acts as symmetries of the surface**.

Since the Cayley graph already has a natural left action of G ,

$$g \cdot x = gx,$$

this symmetry should extend to the whole surface.

Therefore the **faces must also be preserved by the action of$G$ **.

> **Key Principle:**
> The set of surface cycles must be G **-invariant**.

This means that if a cycle is used as a face, then **all of its G -translates must also be faces**.

General Framework

Step 1. Choose a based cycle

Select a cycle in the Cayley graph starting at the identity:

$$C_1 = (e, g_1, g_2, \dots, g_k, e)$$

This is called a **based cycle**.

Interpret this cycle as the boundary of a potential face.

Step 2. Generate symmetric copies

Because we require G**-invariance**, we must include all translated copies of the cycle:

$$\mathcal{O}(C_1) = \{\, hC_1 \mid h \in G \,\}.$$

These cycles form the first set of candidate faces.

Step 3. Count edge usage

For all cycles in $\mathcal{O}(C_1)$, count how many times each edge of the graph appears.

Three situations may occur.

Case 1 — Success

Every edge appears **exactly twice**.

Then these cycles define a valid set of faces of a closed surface.

Case 2 — Failure

Some edge appears **more than twice**.

This means too many faces meet along that edge, so the construction fails.

Case 3 — Incomplete surface

Every edge appears **at most twice**, but some edges appear only once or not at all.

This means the surface is incomplete and more faces are needed.

Step 4. Add another based cycle

Choose another cycle

$$C_2$$

that is not already included.

Again we must include all of its symmetric copies:

$$\mathcal{O}(C_2).$$

Add these cycles to the face set and repeat the counting process.

Step 5. Continue the process

Add cycles

$$C_3, C_4, \dots$$

until either

- every edge appears **exactly twice** (success), or
- some edge appears **more than twice** (failure).

Important Observation

Because Cayley graphs are symmetric, it is usually enough to check **edges incident to the identity vertex**.

If the condition holds there, it holds everywhere.

Example: The Prism from D_n

Consider the dihedral group

$$D_n = \langle r, s \mid r^n = 1,\ s^2 = 1,\ srs = r^{-1} \rangle$$

and the Cayley graph

$$\Gamma = \mathrm{Cay}(D_n, \{r, r^{-1}, s\}).$$

This graph is the **1-skeleton of the standard n-prism**.

Step 1: The n-cycles

Consider the cycle

$$C_1 = (e, r, r^2, \ldots, r^{n-1}, e).$$

Its D_n-orbit produces two n-gons:

- the top face
- the bottom face

Step 2: The 4-cycles

Consider the cycle

$$C_2 = (e, r, sr, s, e).$$

Its D_n-orbit produces n cycles of length 4.

These correspond to the rectangular side faces of the prism.

Step 3: Edge counting

Check how edges are used.

r-edges

Each r-edge appears

- once in an n-gon
- once in a rectangle.

s-edges

Each s-edge appears in **two rectangles**.

Thus every edge appears **exactly twice**.

Result

The cycles from C_1 and C_2 form a D_n-invariant face structure.

Therefore the resulting surface admits D_n **as a symmetry group**.

Euler Check

Vertices:

$$V = 2n$$

Edges:

$$E = 3n$$

Faces:

$$F = n + 2$$

Thus

$$V - E + F = 2n - 3n + (n+2) = 2$$

so the surface is a **sphere**, as expected for a prism.

Advice for the Project

1. Start with **small groups** such as
   D_4, D_5, or A_4.

2. Draw the **Cayley graph** first.

3. Express cycles as **words in generators**, e.g.

$$s_1 s_2 \cdots s_k = 1$$

4. Always include **all G-translates** of a cycle.

5. Verify the key condition:

> each edge must belong to **exactly two faces**.

Explicit Example: $G = D_4$ and a Based 6-Cycle

Let

$$D_4 = \langle r, s \mid r^4 = 1,\ s^2 = 1,\ srs = r^{-1} \rangle.$$

We consider the Cayley graph

$$\Gamma = \mathrm{Cay}(D_4, \{r, r^{-1}, s\}).$$

Its vertex set is

$$D_4 = \{1, r, r^2, r^3, s, rs, r^2s, r^3s\}.$$

This graph is the usual cube graph.

Step 1. Start with a based cycle of length 6

Consider the cycle

$$C = (1,\ r,\ r^2,\ r^2s,\ rs,\ s,\ 1).$$

We first check that this is indeed a cycle in the Cayley graph.

The successive edges are

- $1 \to r$ by multiplying by r,
- $r \to r^2$ by multiplying by r,
- $r^2 \to r^2s$ by multiplying by s,
- $r^2s \to rs$ by multiplying by r,
- $rs \to s$ by multiplying by r,
- $s \to 1$ by multiplying by s.

So the corresponding word is

$$r \cdot r \cdot s \cdot r \cdot r \cdot s.$$

Using $srs = r^{-1}$, we get

$$r^2sr^2s = r^2r^{-2}s^2 = 1,$$

so this path really closes up.

Step 2. Take the D_4-orbit

Because we want the resulting face set to be D_4-invariant, we must include all left translates of C.

Starting from some representatives, we obtain the following four distinct hexagons:

$$F_0 = (1,\ r,\ r^2,\ r^2s,\ rs,\ s,\ 1),$$

$$F_1 = (r,\ r^2,\ r^3,\ r^3s,\ r^2s,\ rs,\ r),$$

$$F_2 = (r^2,\ r^3,\ 1,\ s,\ r^3s,\ r^2s,\ r^2),$$

$$F_3 = (r^3,\ 1,\ r,\ rs,\ s,\ r^3s,\ r^3).$$

These are all obtained from F_0 by left multiplication by $1, r, r^2, r^3$.

Some other translates, such as sF_0, do not give new faces; they are the same hexagons with the opposite orientation. So the orbit gives exactly 4 distinct faces.

Step 3. Check all edges of the cube

The Cayley graph has 8 vertices and 12 edges.

We now list the edges appearing in the four faces.

Edges of F_0

$$\{1,r\},\ \{r,r^2\},\ \{r^2,r^2s\},\ \{r^2s,rs\},\ \{rs,s\},\ \{s,1\}.$$

Edges of F_1

$$\{r,r^2\},\ \{r^2,r^3\},\ \{r^3,r^3s\},\ \{r^3s,r^2s\},\ \{r^2s,rs\},\ \{rs,r\}.$$

Edges of F_2

$$\{r^2,r^3\},\ \{r^3,1\},\ \{1,s\},\ \{s,r^3s\},\ \{r^3s,r^2s\},\ \{r^2s,r^2\}.$$

Edges of F_3

$$\{r^3,1\},\ \{1,r\},\ \{r,rs\},\ \{rs,s\},\ \{s,r^3s\},\ \{r^3s,r^3\}.$$

Step 4. Count how many times each edge appears

Now we group the edges and record in which faces they appear.

$$\{1, r\} \text{ appears in } F_0, F_3,$$

$$\{r, r^2\} \text{ appears in } F_0, F_1,$$

$$\{r^2, r^3\} \text{ appears in } F_1, F_2,$$

$$\{r^3, 1\} \text{ appears in } F_2, F_3,$$

$$\{1, s\} \text{ appears in } F_0, F_2,$$

$$\{r, rs\} \text{ appears in } F_1, F_3,$$

$$\{r^2, r^2 s\} \text{ appears in } F_0, F_2,$$

$$\{r^3, r^3 s\} \text{ appears in } F_1, F_3,$$

$$\{s, rs\} \text{ appears in } F_0, F_3,$$

$$\{rs, r^2 s\} \text{ appears in } F_0, F_1,$$

$$\{r^2 s, r^3 s\} \text{ appears in } F_1, F_2,$$

$$\{r^3 s, s\} \text{ appears in } F_2, F_3.$$

So **every edge appears exactly twice**.

This is exactly the condition we want.

Step 5. Conclusion

The D_4-orbit of the single based cycle

$$(1, r, r^2, r^2 s, rs, s, 1)$$

already gives a complete D_4-invariant face structure on the Cayley graph.

Therefore the cube graph can be regarded as the 1-skeleton of a closed surface with 4 hexagonal faces.

Step 6. Euler characteristic

We have

$$V = 8, \qquad E = 12, \qquad F = 4.$$

Hence

$$\chi = V - E + F = 8 - 12 + 4 = 0.$$

So the resulting closed surface has Euler characteristic 0, hence it is a torus.

Final remark for students

This example is important because it shows that the same Cayley graph of D_4 can support different D_4-invariant face structures:

- the usual prism structure uses two 4-cycles and four side 4-cycles, giving the sphere;
- the single orbit of the above 6-cycle gives four hexagonal faces, giving the torus.

So the same group and the same Cayley graph may lead to different closed surfaces, depending on which G-invariant cycle system we choose.